

ULTIMATE REASONING COMPENDIUM

Comprehensive Theory, Step-by-Step Examples & 25 High-Yield Practice Questions
Across All Major Topics

Table of Contents

- Topic 1: Number Series
- Topic 2: Coding & Decoding
- Topic 3: Blood Relations
- Topic 4: Direction Sense Test
- Topic 5: Analogy
- Topic 6: Classification (Odd One Out)
- Topic 7: Syllogism
- Topic 8: Seating Arrangement
- Topic 9: Puzzles
- Topic 10: Clocks & Calendars

Topic 1: Number Series

1. Theory & Comprehensive Concepts

Number series questions evaluate your logical pattern recognition skills. A series of numbers is given following a specific mathematical rule or logical sequence, and you are required to find the missing or wrong term.

Primary Patterns to Master:

- **Arithmetic Series:** Constant difference between consecutive terms (AP).
- **Geometric Series:** Constant ratio/multiplier between terms (GP).
- **Difference of Differences:** Second or third level differences form an arithmetic or geometric sequence.
- **Square & Cube Series:** Numbers close to squares ($n^2 \pm c$) or cubes ($n^3 \pm c$).
- **Mixed / Alternating Series:** Two alternating patterns interlaced within a single series.

Solved Example: Find the missing term in the series: 3, 7, 15, 31, 63, ?

Solution: Look at differences: $7-3=4$, $15-7=8$, $31-15=16$, $63-31=32$. Differences form a geometric progression multiplying by 2 (4, 8, 16, 32). Next difference should be 64.

$$63 + 64 = 127.$$

Answer: 127

Practice Questions (25 Questions)

1. Find the missing number in the series: 2, 5, 10, 17, 26, ?

- | | |
|--------|--------|
| (A) 35 | (B) 37 |
| (C) 39 | (D) 41 |

2. Find the next term: 4, 9, 19, 39, 79, ?

- | | |
|---------|---------|
| (A) 159 | (B) 149 |
| (C) 139 | (D) 169 |

3. Find the missing term: 6, 12, 20, 30, 42, ?

- | | |
|--------|--------|
| (A) 54 | (B) 56 |
| (C) 58 | (D) 60 |

4. Complete the series: 5, 10, 20, 40, 80, ?

(A) 120

(B) 140

(C) 160

(D) 180

5. Find the missing term: 2, 3, 5, 7, 11, 13, ?

(A) 15

(B) 17

(C) 19

(D) 21

6. Find the wrong term in 3, 8, 15, 24, 35, 48, 63, 80, 99... Wait, let's look at: 1, 4, 9, 16, 25, 36, 49, 64, 81, ?

(A) 100

(B) 121

(C) 144

(D) 96

7. Complete the series: 0, 2, 6, 12, 20, 30, ?

(A) 40

(B) 42

(C) 44

(D) 46

8. Find the missing number: 3, 9, 27, 81, ?

(A) 162

(B) 243

(C) 324

(D) 729

9. Find the next term: 8, 24, 12, 36, 18, 54, ?

(A) 27

(B) 32

(C) 42

(D) 64

10. Complete the series: 1, 8, 27, 64, 125, ?

(A) 196

(B) 216

(C) 225

(D) 256

11. Find the missing term: 10, 14, 26, 42, 70, ?

(A) 100

(B) 104

(C) 108

(D) 112

12. Find the missing term: 2, 6, 12, 20, 30, 42, 56, ?

(A) 70

(B) 72

(C) 74

(D) 76

13. Complete the series: 120, 99, 80, 63, 48, ?

(A) 35

(B) 36

(C) 39

(D) 41

14. Find the missing number: 4, 5, 7, 10, 14, 19, ?

(A) 24

(B) 25

(C) 26

(D) 27

15. Find the next term: 2, 3, 6, 7, 14, 15, ?

(A) 30

(B) 28

(C) 31

(D) 29

16. Complete the series: 24, 60, 120, 210, ?

(A) 300

(B) 336

(C) 350

(D) 380

17. Find the missing number: 1, 2, 4, 7, 11, 16, ?

(A) 21

(B) 22

(C) 23

(D) 24

18. Find the next term: 3, 6, 18, 72, 360, ?

(A) 1800

(B) 2160

(C) 2520

(D) 1440

19. Complete the series: 8, 28, 116, 584, ?

(A) 2500

(B) 3510

(C) 3512

(D) 3516

20. Find the missing term: 7, 10, 8, 11, 9, 12, ?

(A) 10

(B) 13

(C) 14

(D) 15

21. Find the next term: 1, 4, 27, 256, ?

(A) 3125

(B) 1024

(C) 625

(D) 4096

22. Complete the series: 10, 22, 46, 94, 190, ?

- | | |
|---------|---------|
| (A) 380 | (B) 382 |
| (C) 384 | (D) 386 |

23. Find the missing term: 15, 31, 63, 127, 255, ?

- | | |
|---------|---------|
| (A) 511 | (B) 513 |
| (C) 515 | (D) 517 |

24. Find the next term: 2, 15, 41, 80, 132, ?

- | | |
|---------|---------|
| (A) 195 | (B) 197 |
| (C) 199 | (D) 201 |

25. Complete the series: 1, 3, 9, 31, 129, ?

- | | |
|---------|---------|
| (A) 651 | (B) 653 |
| (C) 655 | (D) 657 |

Answer Key - Topic 1: Number Series

1: (B) | 2: (A) | 3: (B) | 4: (C) | 5: (B) | 6: (A) | 7: (B) | 8: (B) | 9:
(A) | 10: (B)

11: (B) | 12: (B) | 13: (A) | 14: (B) | 15: (A) | 16: (B) | 17: (A) | 18: (B)
| 19: (D) | 20: (A)

21: (A) | 22: (B) | 23: (A) | 24: (B) | 25: (A)

Topic 2: Coding & Decoding

1. Theory & Comprehensive Concepts

Coding and Decoding is a method used to encrypt words, numbers, or sentences into secret codes based on specific rules. Mastery depends on knowing alphabetical letter positions (A=1 to Z=26) and reverse pairs (A=Z, B=Y, etc.).

Primary Coding Patterns:

- **Forward/Backward Shifting:** Adding or subtracting fixed or progressive numbers from letter positions.
- **Opposite Letter Coding:** Replacing letters with their alphabetical opposites.
- **Number/Symbol Coding:** Assigning specific numbers or symbols to letters.
- **Conditional Coding:** Applying specific rules based on vowels/consonants.

Solved Example: If 'FLOWER' is coded as 'GMNXFS', how is 'GARDEN' coded?

Solution: Each letter is shifted forward by +1 position.

F→G, L→M, O→N, W→X, E→F, R→S.

Applying to GARDEN: G→H, A→B, R→S, D→E, E→F, N→O.

Answer: HBSEFO

Practice Questions (25 Questions)

1. If 'CAT' is coded as 'DBU', how is 'DOG' coded?

- | | |
|---------|---------|
| (A) EPH | (B) FPH |
| (C) EQH | (D) DPH |

2. If 'PENCIL' is written as 'QFOdJM', how is 'PAPER' written?

- | | |
|-----------|-----------|
| (A) QBQFS | (B) QBPFS |
| (C) RAQFS | (D) PBQER |

3. If 'WATER' is coded as 'XBUFS', how is 'FIRE' coded?

- | | |
|----------|----------|
| (A) GJSF | (B) GJSE |
| (C) HJTF | (D) FKQD |

4. If 'ROSE' is coded as '6821', 'CHAIR' is coded as '73456', what is code for 'SEARCH'?
- (A) 214673 (B) 214763
(C) 216473 (D) 214637
5. If 'RED' = 27, then 'PEN' = ?
- (A) 35 (B) 34
(C) 36 (D) 38
6. If 'MANGO' is coded as 'NZMTV', how is 'APPLE' coded?
- (A) ZKKOV (B) ZKKOV
(C) YJJOV (D) ZJKOV
7. If 'GOOD' is coded as 'HPPE', how is 'BEST' coded?
- (A) CFTU (B) CFSU
(C) BFTU (D) CGTU
8. If 'STOVE' is written as 'FNTSU', how is 'CREAM' written?
- (A) BQZDL (B) BLZQD
(C) BQZLD (D) BNZDL
9. If 'ZEBRA' = 'AFCSB', then 'LION' = ?
- (A) MJPO (B) MKPO
(C) NKPO (D) MJOP
10. If 'ROAD' is coded as 'URDG', how is 'SWAN' coded?
- (A) VZDQ (B) VZDQ
(C) VZBQ (D) VZCQ
11. If 'BAT' = 23, 'CAT' = 24, then 'BALL' = ?
- (A) 27 (B) 28
(C) 29 (D) 30
12. If 'TIGER' is coded as 'QDFHS', how is 'FISH' coded?
- (A) CHRG (B) CGRH
(C) DHRG (D) CHGR

13. If 'COMPUTER' is coded as 'RFUVQNPC', how is 'MEDICINE' coded?
- (A) EOJDJEFM (B) EOJDEJFM
(C) EOJDJFME (D) EOJDJEMF
14. If 'SUN' = 54, 'MOON' = 88, then 'STAR' = ?
- (A) 75 (B) 76
(C) 77 (D) 78
15. If 'WHITE' is called 'BLACK', 'BLACK' is called 'RED', 'RED' is called 'BLUE', what is the color of clear sky?
- (A) Blue (B) Red
(C) Black (D) White
16. If 'DELHI' = 'CCKGD', then 'MUMBAI' = ?
- (A) LTLAZG (B) LTLAZH
(C) LTMAZG (D) KSLAZG
17. If 'HOT' = 'ILW', how is 'COLD' coded?
- (A) FROG (B) FQOG
(C) FRQG (D) FRPG
18. If 'TRAIN' = 'WUDLQ', then 'BUS' = ?
- (A) EXV (B) EWV
(C) FXV (D) EYV
19. If 'PEN' = '11', 'NOTE' = '15', then 'BOOK' = ?
- (A) 12 (B) 13
(C) 14 (D) 15
20. If 'JUMP' = 'KVOQ', how is 'WALK' coded?
- (A) XBMV (B) XBLZ
(C) XBMZ (D) YBMZ
21. If 'GARDEN' is coded as 'ITUFGP', how is 'FLOWER' coded?
- (A) HNQYGT (B) HNPXGT
(C) HNQYFS (D) HNQXGT

22. If 'BOOK' = '43', how is 'PEN' coded?

(A) 35

(B) 34

(C) 36

(D) 37

23. If 'FATHER' = 'DBVFCQ', how is 'MOTHER' coded?

(A) KMRFCQ

(B) KMRFCP

(C) LMRFCQ

(D) KMSFCQ

24. If 'STARS' = 'TRBST', how is 'MOONS' coded?

(A) NPNPT

(B) NPONT

(C) NPOOT

(D) NQNPT

25. If 'LION' = '1291514', how is 'TIGER' coded?

(A) 2097518

(B) 2097618

(C) 2098518

(D) 1997518

Answer Key - Topic 2: Coding & Decoding

1: (A) | 2: (A) | 3: (A) | 4: (A) | 5: (A) | 6: (A) | 7: (A) | 8: (A) | 9:
(A) | 10: (A)

11: (A) | 12: (A) | 13: (A) | 14: (A) | 15: (B) | 16: (A) | 17: (A) | 18: (A)
| 19: (A) | 20: (C)

21: (A) | 22: (A) | 23: (A) | 24: (A) | 25: (A)

Topic 3: Blood Relations

1. Theory & Comprehensive Concepts

Blood Relations test family linkage analysis. To solve effortlessly, use gender symbols (+ for male, - for female), horizontal lines for spouses, and vertical lines for generations.

Key Generational Links:

- Father's/Mother's father = Grandfather | Mother = Grandmother
- Brother's/Sister's son = Nephew | Daughter = Niece
- Father's/Mother's brother = Uncle | Sister = Aunt
- Son's wife = Daughter-in-law | Daughter's husband = Son-in-law

Solved Example: Pointing to a photograph, Rohit said, "She is the daughter of my grandfather's only son." How is the girl related to Rohit?

Solution: Rohit's grandfather's only son is Rohit's father. The girl is the daughter of Rohit's father, which means she is Rohit's sister.

Answer: Sister

Practice Questions (25 Questions)

1. Pointing to a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?
(A) Mother (B) Sister
(C) Aunt (D) Grandmother
2. A is B's brother. C is A's mother. D is C's father. E is D's mother. How is A related to D?
(A) Grandson (B) Son
(C) Grandfather (D) Great-Grandson
3. If $P + Q$ means P is the mother of Q, $P - Q$ means P is the brother of Q, which of the following means 'A is the maternal uncle of B'?
(A) $A - C + B$ (B) $A + C - B$
(C) $A - B + C$ (D) $C - A + B$

4. Introducing a boy, a girl said, "He is the son of the daughter of the father of my uncle." How is the boy related to the girl?
- (A) Brother (B) Cousin
(C) Nephew (D) Uncle
5. If X is the brother of the son of Y's son, how is X related to Y?
- (A) Son (B) Grandson
(C) Great-Grandson (D) Nephew
6. Pointing to a photograph, a person says, "The man in the photo is the son of my father's sister." Who is the person to the man?
- (A) Cousin (B) Nephew
(C) Uncle (D) Brother
7. A and B are young brothers. E is the father of D. B is the brother of C. C is the daughter of D. Who is the father of B?
- (A) A (B) E
(C) C (D) Cannot be determined
8. If M is the sister of Z, Z is the husband of P, P is the daughter of A, how is M related to A?
- (A) Daughter (B) Sister
(C) Daughter-in-law (D) Sister-in-law
9. If $A + B$ means father of B; $A - B$ means wife of B; $A * B$ means brother of B, which means 'D is son of C'?
- (A) $C + D * E$ (B) $C - D + E$
(C) $C * D + E$ (D) $C + D - E$
10. Pointing to a lady, Ramesh said, "Her daughter is the only daughter of my wife." How is Ramesh related to the lady?
- (A) Son (B) Son-in-law
(C) Husband (D) Father
11. K is the mother of R. R is the sister of J. J is the father of S. How is K related to S?
- (A) Grandmother (B) Mother
(C) Aunt (D) Great-Grandmother

12. If P is the husband of Q and R is the mother of Q and S, how is P related to R?
- (A) Son (B) Son-in-law
(C) Brother-in-law (D) Nephew
13. A woman introduces a man as the son of the brother of her mother. How is the man related to the woman?
- (A) Cousin (B) Nephew
(C) Son (D) Uncle
14. If A * B means brother; A / B means mother, which shows M is maternal uncle of T?
- (A) M * K / T (B) M / K * T
(C) M * T / K (D) T / K * M
15. Pointing to a girl in the park, Akash said, "She is the daughter of my mother's only son." How is the girl related to Akash?
- (A) Daughter (B) Sister
(C) Niece (D) Cousin
16. If X is the wife of Y and Y is the brother of Z, how is Z related to X if Z is female?
- (A) Sister-in-law (B) Sister
(C) Mother-in-law (D) Aunt
17. In a family, A, B, C, D, E, F. A and B married. A is male. C is only son of B. E is sister of C. B is sister of D. Who is mother of C?
- (A) A (B) B
(C) D (D) E
18. If A % B means son; A & B means daughter, which means X is granddaughter of Z?
- (A) X & Y % Z (B) X % Y & Z
(C) X % Y % Z (D) X & Y & Z
19. Pointing to a photograph of a boy, Sunita said, "He is the son of the only daughter of my grandfather." How is the boy related to Sunita?
- (A) Brother (B) Cousin
(C) Uncle (D) Son

20. If D is the brother of B, how is B related to C if B is the father of C?
- (A) Father (B) Uncle
(C) Son (D) Brother
21. A is father of C and D is son of B. E is brother of A. If C is sister of D, how is B related to E?
- (A) Sister-in-law (B) Brother-in-law
(C) Sister (D) Daughter-in-law
22. If P is Q's brother, R is Q's mother, S is R's father, T is S's mother, how is P related to S?
- (A) Grandson (B) Son
(C) Grandfather (D) Great-Grandson
23. If X is sister of Y, Y is brother of Z, Z is husband of P, how is P related to Y?
- (A) Sister-in-law (B) Wife
(C) Sister (D) Daughter-in-law
24. If A is mother of B, B is sister of C, C is father of D, how is A related to D?
- (A) Grandmother (B) Mother
(C) Aunt (D) Great-Grandmother
25. Pointing to a man, a woman said, "His wife is the only daughter of my mother." Who is the woman to the man?
- (A) Wife (B) Sister
(C) Mother (D) Mother-in-law

Answer Key - Topic 3: Blood Relations

1: (A) | 2: (A) | 3: (A) | 4: (A) | 5: (B) | 6: (A) | 7: (B) | 8: (D) | 9:
(A) | 10: (C)
11: (A) | 12: (B) | 13: (A) | 14: (A) | 15: (A) | 16: (A) | 17: (B) | 18: (A)
| 19: (A) | 20: (A)
21: (A) | 22: (A) | 23: (A) | 24: (A) | 25: (A)

Topic 4: Direction Sense Test

1. Theory & Comprehensive Concepts

Direction Sense test questions evaluate your tracking of movements across cardinal directions (North, South, East, West) and diagonal directions, alongside calculating distances using the Pythagoras Theorem ($\text{Hypotenuse}^2 = \text{Base}^2 + \text{Perpendicular}^2$).

Core Rules:

- Top is North, Bottom is South, Right is East, Left is West.
- Right turn = Clockwise 90-degree turn. Left turn = Anti-clockwise 90-degree turn.

Solved Example: A man walks 5 km North, turns right and walks 4 km, turns right again and walks 5 km. Which direction is he facing now and how far is he from the starting point?

Solution: He walked North 5km, turned East 4km, turned South 5km. His final position is directly East of his starting point at a distance of 4 km.

Answer: East, 4 km

Practice Questions (25 Questions)

1. A man walks 30m North, turns right and walks 40m, turns right and walks 30m. How far is he from start?
(A) 40 meters (B) 30 meters
(C) 50 meters (D) 70 meters
2. Karan walked 15m East, turned left and walked 15m, turned left and walked 15m. Distance from start?
(A) 15 meters (B) 30 meters
(C) 45 meters (D) 20 meters
3. A dog runs 20m East, turns right 10m, right 9m, left 5m, left 12m. Direction facing now?
(A) North (B) South
(C) East (D) West

4. If South-East becomes North, North-East becomes West, what will West become?
- (A) North-East (B) South-East
(C) South-West (D) North-West
5. A person goes 10 km North, 6 km South, 3 km East. Distance and direction from start?
- (A) 5 km North-East (B) 7 km North-East
(C) 5 km South-East (D) 4 km East
6. Rahul walked 12m North, West 15m, South 12m, West 5m. Distance from start?
- (A) 20 m (B) 15 m
(C) 17 m (D) 22 m
7. Rohan walks 10 km North, 6 km South, 3 km East. Distance and direction from start?
- (A) 5 km North-East (B) 4 km North-East
(C) 7 km East (D) 3 km North
8. If X is south of Y and Z is east of Y, direction of X with respect to Z?
- (A) South-West (B) South-East
(C) North-East (D) North-West
9. A watch reads 4:30. Minute hand points East. Direction of hour hand?
- (A) North-East (B) South-East
(C) North-West (D) South-West
10. Morning after sunrise, Suresh faced a pole. Shadow fell to his right. Direction Suresh faced?
- (A) South (B) North
(C) East (D) West
11. Facing West, turns 45 deg clockwise, 180 deg clockwise, 270 deg anti-clockwise. Facing now?
- (A) South-West (B) West
(C) South (D) South-East
12. Deepak walks 75m East, left 25m, left 40m, left 25m. Distance from start?
- (A) 25 meters (B) 35 meters
(C) 50 meters (D) 40 meters

13. Bus A runs 20km North, right 25km. Bus B runs 20km East, right 5km. Bus A w.r.t. Bus B?
- (A) 20 km North (B) 25 km North-West
(C) 20 km North-West (D) 15 km North
14. Boy rode North, left 1km, left 2km, found 1km west of start. Initial northward ride?
- (A) 1 km (B) 2 km
(C) 3 km (D) 5 km
15. Maya walks 4km West, left 3km. Distance from start?
- (A) 5 km (B) 7 km
(C) 1 km (D) 4 km
16. If North-West becomes East, North-East becomes South, what does East become?
- (A) North-West (B) South-West
(C) South-East (D) North-East
17. Man walks 5km South, right 3km, left 5km. Direction from start?
- (A) South-West (B) North-East
(C) South-East (D) West
18. Amit walked 10m East, right 10m, left 10m, left 10m. Distance from start?
- (A) 20 m (B) 10 m
(C) 30 m (D) 40 m
19. Postman walked 20m, right 10m, right 20m, left 25m. Distance from office?
- (A) 35 m (B) 25 m
(C) 30 m (D) 45 m
20. Facing North-East, turn 135 degrees clockwise. Direction faced?
- (A) South (B) East
(C) West (D) South-East
21. Person walks 4km West, right 4km, right 4km. Distance from start?
- (A) 4 km (B) 8 km
(C) 0 km (D) 12 km

22. Place P, walked 10m North, East 20m, South 10m, West 10m. Where now w.r.t P?

(A) 10m East

(B) 10m West

(C) 20m East

(D) At P

23. Vimal walks 5km North, 3km West, 5km South. Distance from start?

(A) 3 km

(B) 5 km

(C) 8 km

(D) 2 km

24. A is 50m north of B, C is 50m east of B. Direction of C w.r.t A?

(A) South-East

(B) North-East

(C) South-West

(D) North-West

25. Clock at 12 noon minute hand points North-East. Hour hand at 1:30 PM points?

(A) South-East

(B) North-West

(C) South-West

(D) North-East

Answer Key - Topic 4: Direction Sense Test

1: (A) | 2: (A) | 3: (A) | 4: (B) | 5: (A) | 6: (A) | 7: (A) | 8: (A) | 9:
(A) | 10: (A)

11: (A) | 12: (B) | 13: (A) | 14: (B) | 15: (A) | 16: (A) | 17: (A) | 18: (A)
| 19: (A) | 20: (A)

21: (A) | 22: (A) | 23: (A) | 24: (A) | 25: (A)

Topic 5: Analogy

1. Theory & Comprehensive Concepts

Analogy tests the ability to discover a common relationship between a given pair of items and apply that exact relationship to find a missing pair. Analogies span across words, numbers, and letters.

Common Analogy Relationships:

- Synonyms & Antonyms | Tool and Object | Worker and Tool
- Country and Capital/Currency | Cause and Effect | Study and Topic

Solved Example: Doctor : Patient :: Teacher : ?

Solution: A doctor treats/guides a patient. Similarly, a teacher teaches/guides a student.

Answer: Student

Practice Questions (25 Questions)

1. Book : Reading :: Fork : ?

- | | |
|-------------|-------------|
| (A) Drawing | (B) Eating |
| (C) Cooking | (D) Writing |

2. Ocean : Water :: Glacier : ?

- | | |
|------------------|----------|
| (A) Refrigerator | (B) Ice |
| (C) Mountain | (D) Cave |

3. Clock : Time :: Thermometer : ?

- | | |
|-----------------|---------------|
| (A) Heat | (B) Radiation |
| (C) Temperature | (D) Energy |

4. Car : Garage :: Aeroplane : ?

- | | |
|------------|------------|
| (A) Port | (B) Depot |
| (C) Hangar | (D) Harbor |

5. Lion : Den :: Eskimo : ?

(A) Tent

(B) Igloo

(C) Cave

(D) House

6. Architect : Building :: Sculptor : ?

(A) Museum

(B) Stone

(C) Statue

(D) Chisel

7. India : Rupee :: USA : ?

(A) Dollar

(B) Yen

(C) Euro

(D) Peso

8. Safe : Secure :: Dangerous : ?

(A) Bold

(B) Risky

(C) Fear

(D) Sure

9. Flow : River :: Stagnant : ?

(A) Rain

(B) Stream

(C) Pool

(D) Canal

10. Bird : Fly :: Snake : ?

(A) Crawl

(B) Walk

(C) Run

(D) Jump

11. 7 : 48 :: 11 : ?

(A) 120

(B) 122

(C) 121

(D) 119

12. 8 : 64 :: 9 : ?

(A) 72

(B) 81

(C) 90

(D) 99

13. 25 : 125 :: 36 : ?

(A) 216

(B) 180

(C) 200

(D) 196

14. CF : HX :: EJ : ?

(A) JQ

(B) KP

(C) IR

(D) LS

15. BDF : HJL :: NPR : ?

(A) TVX

(B) UWY

(C) SUW

(D) RTW

16. 12 : 30 :: 20 : ?

(A) 35

(B) 42

(C) 50

(D) 56

17. AZ : BY :: CX : ?

(A) DW

(B) EV

(C) FU

(D) GT

18. Meter : Length :: Scale : ?

(A) Weight

(B) Measurement

(C) Time

(D) Mass

19. Oxygen : Burn :: Carbon Dioxide : ?

(A) Isolate

(B) Foam

(C) Extinguish

(D) Explode

20. Pen : Writer :: Needle : ?

(A) Tailor

(B) Carpenter

(C) Doctor

(D) Sculptor

21. 11 : 121 :: 15 : ?

(A) 215

(B) 225

(C) 235

(D) 245

22. Drama : Director :: Magazine : ?

(A) Editor

(B) Author

(C) Publisher

(D) Reader

23. Good : Bad :: Roof : ?

(A) Wall

(B) Floor

(C) Pillar

(D) Foundation

24. ACE : GIK :: MOQ : ?

(A) SUW

(B) STV

(C) RTW

(D) TVX

25. 14 : 9 :: 26 : ?

(A) 15

(B) 16

(C) 17

(D) 18

Answer Key - Topic 5: Analogy

1: (B) | 2: (B) | 3: (C) | 4: (C) | 5: (B) | 6: (C) | 7: (A) | 8: (B) | 9: (C) | 10: (A)

11: (B) | 12: (B) | 13: (A) | 14: (A) | 15: (A) | 16: (B) | 17: (A) | 18: (B) | 19: (C) | 20: (A)

21: (B) | 22: (A) | 23: (B) | 24: (A) | 25: (A)

Topic 6: Classification (Odd One Out)

1. Theory & Comprehensive Concepts

Classification requires identifying four or five given options, among which all except one share a common characteristic or property. The task is to find the odd item out.

Basis of Classification:

- Alphabetical sequence/gaps | Prime vs Composite numbers | Square/Cube properties
- Semantic categories (Animals, Professions, Elements, etc.)

Solved Example: Find the odd one out: (A) Carrot, (B) Potato, (C) Onion, (D) Apple

Solution: Carrot, Potato, and Onion grow underground (roots/tubers), whereas Apple grows on a tree.

Answer: (D) Apple

Practice Questions (25 Questions)

1. Find the odd one out: (A) Iron, (B) Aluminum, (C) Copper, (D) Brass

- | | |
|------------|--------------|
| (A) Iron | (B) Aluminum |
| (C) Copper | (D) Brass |

2. Find the odd one out: (A) 27, (B) 64, (C) 125, (D) 144

- | | |
|---------|---------|
| (A) 27 | (B) 64 |
| (C) 125 | (D) 144 |

3. Find the odd one out: (A) Dog, (B) Cat, (C) Lion, (D) Cow

- | | |
|----------|---------|
| (A) Dog | (B) Cat |
| (C) Lion | (D) Cow |

4. Find the odd one out: (A) March, (B) May, (C) July, (D) November

- | | |
|-----------|--------------|
| (A) March | (B) May |
| (C) July | (D) November |

5. Find the odd one out: (A) 11, (B) 13, (C) 15, (D) 17

(A) 11

(B) 13

(C) 15

(D) 17

6. Find the odd one out: (A) BD, (B) IK, (C) PR, (D) TV

(A) BD

(B) IK

(C) PR

(D) TV

7. Find the odd one out: (A) Ear, (B) Eye, (C) Nose, (D) Heart

(A) Ear

(B) Eye

(C) Nose

(D) Heart

8. Find the odd one out: (A) 81, (B) 64, (C) 49, (D) 32

(A) 81

(B) 64

(C) 49

(D) 32

9. Find the odd one out: (A) Wheat, (B) Rice, (C) Mustard, (D) Barley

(A) Wheat

(B) Rice

(C) Mustard

(D) Barley

10. Find the odd one out: (A) AEI, (B) MNO, (C) UIO, (D) EIO

(A) AEI

(B) MNO

(C) UIO

(D) EIO

11. Find the odd one out: (A) 12 - 144, (B) 13 - 169, (C) 14 - 196, (D) 15 - 230

(A) 12 - 144

(B) 13 - 169

(C) 14 - 196

(D) 15 - 230

12. Find the odd one out: (A) Kilometer, (B) Decimeter, (C) Furlong, (D) Pound

(A) Kilometer

(B) Decimeter

(C) Furlong

(D) Pound

13. Find the odd one out: (A) Tennis, (B) Cricket, (C) Chess, (D) Hockey

(A) Tennis

(B) Cricket

(C) Chess

(D) Hockey

14. Find the odd one out: (A) JKL, (B) PQR, (C) XYZ, (D) DEF
- | | |
|---------|---------|
| (A) JKL | (B) PQR |
| (C) XYZ | (D) DEF |
15. Find the odd one out: (A) 35, (B) 49, (C) 63, (D) 77
- | | |
|--------|--------|
| (A) 35 | (B) 49 |
| (C) 63 | (D) 77 |
16. Find the odd one out: (A) Author, (B) Reader, (C) Publisher, (D) Editor
- | | |
|---------------|------------|
| (A) Author | (B) Reader |
| (C) Publisher | (D) Editor |
17. Find the odd one out: (A) 17, (B) 23, (C) 29, (D) 33
- | | |
|--------|--------|
| (A) 17 | (B) 23 |
| (C) 29 | (D) 33 |
18. Find the odd one out: (A) AZBY, (B) CXDW, (C) EVFU, (D) GSHT
- | | |
|----------|----------|
| (A) AZBY | (B) CXDW |
| (C) EVFU | (D) GSHT |
19. Find the odd one out: (A) Mercury, (B) Venus, (C) Moon, (D) Mars
- | | |
|-------------|-----------|
| (A) Mercury | (B) Venus |
| (C) Moon | (D) Mars |
20. Find the odd one out: (A) Flute, (B) Guitar, (C) Violin, (D) Sitar
- | | |
|------------|------------|
| (A) Flute | (B) Guitar |
| (C) Violin | (D) Sitar |
21. Find the odd one out: (A) 24, (B) 35, (C) 48, (D) 63
- | | |
|--------|--------|
| (A) 24 | (B) 35 |
| (C) 48 | (D) 63 |
22. Find the odd one out: (A) Ship, (B) Boat, (C) Submarine, (D) Glider
- | | |
|---------------|------------|
| (A) Ship | (B) Boat |
| (C) Submarine | (D) Glider |

23. Find the odd one out: (A) 8 - 4, (B) 27 - 9, (C) 64 - 16, (D) 125 - 25

(A) 8 - 4

(B) 27 - 9

(C) 64 - 16

(D) 125 - 25

24. Find the odd one out: (A) CEG, (B) IKM, (C) OQS, (D) TVX

(A) CEG

(B) IKM

(C) OQS

(D) TVX

25. Find the odd one out: (A) January, (B) April, (C) June, (D) September

(A) January

(B) April

(C) June

(D) September

Answer Key - Topic 6: Classification

1: (D) [Brass is an alloy] | 2: (D) | 3: (C) [Wild animal] | 4: (D) [30 days]
| 5: (C) [Composite]

6: (B) | 7: (D) [Internal organ] | 8: (D) | 9: (C) [Oilseed] | 10: (B)

11: (D) | 12: (D) [Weight unit] | 13: (C) [Indoor board game] | 14: (C) [Not sequential]

15: (D) | 16: (B) | 17: (D) | 18: (D) | 19: (C) [Satellite] | 20: (A) [Wind instrument]

21: (B) | 22: (D) [Aviation] | 23: (A) | 24: (D) | 25: (A) [31 days]

Topic 7: Syllogism

1. Theory & Comprehensive Concepts

Syllogism evaluates logical deduction from a given set of statements. Statements are assumed to be true even if they contradict general facts. Solving is best performed using Venn Diagrams.

Standard Categorical Statements:

- **All A are B:** Universal Affirmative.
- **Some A are B:** Particular Affirmative.
- **No A is B:** Universal Negative.
- **Some A are not B:** Particular Negative.

Solved Example: Statements: All cats are dogs. All dogs are animals. Conclusions: I. All cats are animals. II. Some animals are cats.

Solution: Circle of Cats inside Dogs, inside Animals. Both conclusions logically follow.

Answer: Both I and II follow

Practice Questions (25 Questions)

- Statements: All pens are books. All books are pencils. Conclusions: I. All pens are pencils. II. Some pencils are pens.
(A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
- Statements: Some cars are bikes. All bikes are trucks. Conclusions: I. Some cars are trucks. II. All trucks are cars.
(A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
- Statements: No flower is red. All roses are red. Conclusions: I. No rose is flower. II. Some flowers are roses.
(A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows

4. Statements: All doors are windows. Some windows are walls. Conclusions: I. Some doors are walls. II. No door is wall.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Neither follows
5. Statements: All cups are plates. Some plates are spoons. Conclusions: I. Some cups are spoons. II. Some spoons are plates.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
6. Statements: Some pens are pencils. Some pencils are erasers. Conclusions: I. Some pens are erasers. II. No pen is eraser.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Both follow
7. Statements: All trees are plants. All shrubs are plants. Conclusions: I. Some trees are shrubs. II. No tree is shrub.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Both follow
8. Statements: All cats are dogs. No dog is a rat. Conclusions: I. No cat is rat. II. Some rats are cats.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
9. Statements: Some computers are laptops. All laptops are devices. Conclusions: I. Some computers are devices. II. All devices are laptops.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
10. Statements: All birds can fly. No penguin can fly. Conclusions: I. Penguins are not birds. II. Some birds are penguins.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
11. Statements: Some books are novels. All novels are stories. Conclusions: I. Some stories are books. II. All books are stories.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows

12. Statements: No man is mortal. All heroes are mortals. Conclusions: I. No hero is man. II. Some mortals are heroes.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
13. Statements: All boxes are toys. All toys are gifts. Conclusions: I. All boxes are gifts. II. Some gifts are boxes.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
14. Statements: Some stars are moons. Some moons are suns. Conclusions: I. Some stars are suns. II. No star is sun.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Both follow
15. Statements: All papers are files. Some files are folders. Conclusions: I. Some folders are papers. II. No folder is paper.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Neither follows
16. Statements: All pens are markers. All markers are highlighters. Conclusions: I. All highlighters are pens. II. Some highlighters are pens.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
17. Statements: Some keys are locks. Some locks are doors. Conclusions: I. Some keys are doors. II. No key is door.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Both follow
18. Statements: All chairs are tables. Some tables are desks. Conclusions: I. Some desks are chairs. II. No desk is chair.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Neither follows

19. Statements: No pencil is pen. All pens are erasers. Conclusions: I. Some erasers are not pencils.
II. All erasers are pencils.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
20. Statements: All mangoes are fruits. All fruits are sweet. Conclusions: I. All mangoes are sweet. II.
Some sweet are fruits.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
21. Statements: Some dogs are cats. Some cats are bats. Conclusions: I. Some dogs are bats. II. No
dog is bat.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Both follow
22. Statements: All horses are animals. Some animals are wild. Conclusions: I. Some horses are wild.
II. No horse is wild.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Neither follows
23. Statements: All phones are smart. All smart are gadgets. Conclusions: I. All phones are gadgets.
II. Some gadgets are phones.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
24. Statements: Some pens are books. No book is pencil. Conclusions: I. Some pens are not pencils.
II. All pens are pencils.
- (A) Only I follows (B) Only II follows
(C) Both follow (D) Neither follows
25. Statements: All cups are mugs. Some mugs are glasses. Conclusions: I. Some cups are glasses.
II. No cup is glass.
- (A) Only I follows (B) Only II follows
(C) Either I or II follows (D) Neither follows

Answer Key - Topic 7: Syllogism

1: (C) | 2: (A) | 3: (A) | 4: (C) | 5: (B) | 6: (C) | 7: (C) | 8: (A) | 9:
(A) | 10: (A)

11: (A) | 12: (C) | 13: (C) | 14: (C) | 15: (C) | 16: (B) | 17: (C) | 18: (C)
| 19: (A) | 20: (C)

21: (C) | 22: (C) | 23: (C) | 24: (A) | 25: (C)

Topic 8: Seating Arrangement

1. Theory & Comprehensive Concepts

Seating Arrangement tests spatial logic by placing individuals in circular, linear, or rectangular formations facing inwards or outwards based on given conditional statements.

Key Approaches:

- **Linear Arrangement:** Identify extreme ends first; note left/right orientations based on facing direction (North/South).
- **Circular Arrangement:** Facing center implies right is anti-clockwise and left is clockwise. Facing outside reverses this rule.

Solved Example: Five friends A, B, C, D, E sit in a row facing North. A is to the immediate right of B. E is to the left of B but right of C. D is to the right of A. Who is in the middle?

Solution: Order from left to right: C, E, B, A, D. B is in the exact middle.

Answer: B

Practice Questions (25 Questions)

1. Five persons P, Q, R, S, T sit in a row facing North. S is to the immediate left of T, who is at one end. P is to the immediate right of Q. R is between Q and S. Who is in the middle?
(A) Q (B) R
(C) S (D) P
2. Four friends A, B, C, D sit around a circular table facing center. A is opposite C. B is to the right of A. Who is to the left of A?
(A) B (B) C
(C) D (D) Cannot be determined
3. Six people A, B, C, D, E, F sit in a row facing North. A is left of B, C is right of A, D is left of E, F is right of B. Who sits at the extreme left?
(A) A (B) C
(C) D (D) Cannot be determined

4. Five students M, N, O, P, Q sit in a circle facing center. M is between N and O. P is to the immediate left of N. Who is opposite M?
- (A) P (B) Q
(C) N (D) O
5. In a row of 6 boys, A is 3rd from left and B is 4th from right. How many boys between them if total is 6?
- (A) 0 (B) 1
(C) 2 (D) 3
6. Five persons A, B, C, D, E sit in a row facing South. A is to the right of B. E is to the left of C. D is to the left of B. Who is at the right end?
- (A) A (B) B
(C) C (D) D
7. Four persons W, X, Y, Z sit around a square table facing center. W is opposite Y. X is to the immediate right of W. Who is opposite X?
- (A) W (B) Y
(C) Z (D) None
8. Six persons sit in a row. A is adjacent to B and C. D is adjacent to E and F. No one is adjacent to F except D. Who is at the extreme end?
- (A) F (B) A
(C) B (D) C
9. Five people P, Q, R, S, T sit in a row. Q is between P and R. T is to the right of R. S is to the left of P. Who is at the left end?
- (A) S (B) P
(C) Q (D) T
10. Five friends sit in a circle. A is right of B, left of C. D is between C and E. Who is between A and D?
- (A) B (B) C
(C) E (D) None
11. In a row of trees, tree A is 7th from left and 14th from right. Total trees?
- (A) 20 (B) 21
(C) 22

(D) 19

12. Six people M, N, O, P, Q, R sit around a circular table facing center. M is opposite P, N is between M and Q. Who is opposite N?

(A) R

(B) O

(C) Q

(D) P

13. Five boys sit in a row. Rohit is 4th from left and 2nd from right. Total boys?

(A) 5

(B) 6

(C) 7

(D) 4

14. Four players P, Q, R, S sit in a row. P and Q are at ends. R is to the immediate left of Q. Who is 2nd from left?

(A) S

(B) R

(C) Q

(D) P

15. Six students sit in a row facing North. A is to the immediate left of B. C is to the immediate right of D. E is between B and C. Who is at the extreme ends?

(A) A and D

(B) B and C

(C) D and F

(D) A and C

16. Five persons sit in a circle. A is to the right of B and left of E. C is to the left of B. Who is between C and A?

(A) B

(B) E

(C) D

(D) None

17. In a line of 15 people, Aman is 6th from left. Position from right?

(A) 9

(B) 10

(C) 8

(D) 11

18. Four friends W, X, Y, Z sit in a row. X and Y are at center. Z is at left end. Who is at right end?

(A) W

(B) X

(C) Y

(D) Z

19. Five executives sit in a row. CEO is in the middle. Manager is to the immediate left of CEO. Who is at the extreme left if Assistant is at right end?

(A) Clerk

(B) Intern

(C) Officer

(D) Cannot be determined

20. Six people sit around a circular table. A is opposite D, B is opposite E, C is opposite F. Who is to the right of A if facing center?
- (A) B or F (B) C
(C) E (D) D
21. In a row of 20 students, Rahul is 8th from left. Position from right end?
- (A) 12 (B) 13
(C) 14 (D) 11
22. Five children sit in a row. Priya is left of Rahul, right of Amit. Sneha is right of Rahul. Who is in the center?
- (A) Amit (B) Priya
(C) Rahul (D) Sneha
23. Four persons sit around a square table at corners. A faces North-East. Who does he face diagonally?
- (A) South-West (B) North-West
(C) South-East (D) North
24. Six people sit in a row facing North. A and B are at ends. C is immediate left of A. D is immediate right of B. Who is in the middle?
- (A) E and F (B) C and D
(C) A and B (D) None
25. Five members sit in a row. Father is in the middle. Mother is left of Father. Son is right of Mother. Daughter is at extreme right. Who is at extreme left?
- (A) Uncle (B) Grandfather
(C) Father (D) Mother

Answer Key - Topic 8: Seating Arrangement

1: (B) | 2: (C) | 3: (C) | 4: (B) | 5: (A) | 6: (A) | 7: (C) | 8: (A) | 9: (A) | 10: (B)
11: (A) | 12: (B) | 13: (A) | 14: (A) | 15: (A) | 16: (A) | 17: (B) | 18: (A) | 19: (D) | 20: (A)
21: (B) | 22: (C) | 23: (A) | 24: (A) | 25: (B)

Topic 9: Puzzles

1. Theory & Comprehensive Concepts

Puzzle reasoning questions present complex multi-variable conditions (such as matching people with floors, professions, colors, or days). Solving requires constructing a systematic tabular grid to eliminate impossible combinations step by step.

Standard Puzzle Types:

- Floor-based puzzles | Box stacking puzzles | Scheduling / Day-based puzzles | Profession & City matching puzzles.

Solved Example: Three persons A, B, C live on floors 1, 2, and 3 (top). C does not live on an odd floor. A lives below B. Who lives on floor 1?

Solution: C must live on an even floor (floor 2). Since A lives below B, B is on floor 3 and A is on floor 1.

Answer: A

Practice Questions (25 Questions)

1. Three boxes X, Y, Z are stacked one above another. Z is not at the bottom. X is immediately above Y. Which box is at the bottom?
(A) X (B) Y
(C) Z (D) Cannot be determined
2. Four people P, Q, R, S practice different sports: Cricket, Football, Tennis, Hockey. P plays Cricket. R does not play Football or Hockey. Q plays Hockey. What does S play?
(A) Football (B) Tennis
(C) Cricket (D) Hockey
3. Three persons live on a 3-floor building. 3 is top. A lives above B. C lives below B. Who is on floor 2?
(A) A (B) B
(C) C (D) None

4. A, B, C, D have different cars: Audi, BMW, Honda, Toyota. A has Audi. C does not have BMW or Toyota. B has BMW. What does D have?
- (A) Honda (B) Toyota
(C) Audi (D) BMW
5. Five days of week: Mon, Tue, Wed, Thu, Fri. Test on Wed. English is before Math. Math is after Wed. When is English?
- (A) Monday (B) Tuesday
(C) Thursday (D) Friday
6. Three friends have pets: Dog, Cat, Parrot. A does not like birds. B has Cat. What pet does C have?
- (A) Dog (B) Parrot
(C) Cat (D) None
7. Four floors building (1 to 4). P lives on odd floor. Q lives immediately above P. R lives on top floor. Who lives on floor 1?
- (A) P (B) Q
(C) S (D) R
8. Three persons like different colors: Red, Blue, Green. A likes Red. B does not like Green. What does C like?
- (A) Green (B) Blue
(C) Red (D) None
9. Five boxes A, B, C, D, E. C is heavier than D. B is lighter than A. E is heavier than C but lighter than B. Which is heaviest?
- (A) A (B) B
(C) C (D) E
10. Four friends born in different years: 1990, 1992, 1995, 1998. A is oldest. D is youngest. B is older than C. Who was born in 1992?
- (A) B (B) C
(C) D (D) A
11. Three students score different marks: X, Y, Z. X scored more than Y. Z scored less than Y. Who scored lowest?
- (A) X (B) Y
(C) Z (D) Cannot be determined

12. Four cities: Delhi, Mumbai, Pune, Chennai. Delhi is north of Mumbai. Pune is south of Delhi but north of Mumbai. Chennai is south of Mumbai. Which is furthest south?
- (A) Delhi (B) Mumbai
(C) Pune (D) Chennai
13. Three persons: Doctor, Engineer, Teacher. Doctor is older than Engineer. Teacher is younger than Engineer. Who is youngest?
- (A) Doctor (B) Engineer
(C) Teacher (D) None
14. Four books priced differently: P, Q, R, S. P is costlier than Q. R is cheaper than S. Q is costlier than S. Which is cheapest?
- (A) P (B) Q
(C) R (D) S
15. Three houses in a row: Red, Blue, Green. Red is not at ends. Green is to the right of Red. Which house is at left end?
- (A) Red (B) Blue
(C) Green (D) None
16. Four friends have different heights. A is taller than B. C is shorter than D. B is taller than D. Who is tallest?
- (A) A (B) B
(C) C (D) D
17. Three professionals live in cities: X, Y, Z. Architect lives in X. Doctor does not live in Y. Where does Lawyer live if Lawyer is not in Y?
- (A) X (B) Y
(C) Z (D) None
18. Four boxes stacked: 1 to 4. Box 1 is at bottom. Box 4 is at top. Box 2 is immediately above Box 1. Box 3 is below Box 4. Where is Box 3?
- (A) Floor 2 (B) Floor 3
(C) Floor 4 (D) Floor 1
19. Three runners finish a race. A finishes before B. C finishes after B. Who wins?
- (A) A (B) B
(C) C (D) Tie

20. Four students secure ranks 1 to 4. P is rank 1. Q is rank 4. R is higher than S. What is R's rank?
- (A) 2 (B) 3
(C) 2 or 3 (D) Cannot be determined
21. Three people own pets: Dog, Cat, Fish. A owns Dog. B does not own Fish. What does C own?
- (A) Cat (B) Fish
(C) Dog (D) None
22. Four brothers: W, X, Y, Z. W is older than X. Y is younger than Z. X is older than Y. Who is oldest?
- (A) W (B) X
(C) Y (D) Z
23. Three cars: Red, Blue, White. Red is faster than Blue. White is slower than Blue. Fastest car?
- (A) Red (B) Blue
(C) White (D) None
24. Four friends drink: Tea, Coffee, Juice, Milk. A drinks Tea. B does not drink Coffee or Milk. C drinks Coffee. What does D drink?
- (A) Juice (B) Milk
(C) Coffee (D) Tea
25. Three employees: X, Y, Z earn different salaries. X earns more than Y. Z earns less than Y. Who earns maximum?
- (A) X (B) Y
(C) Z (D) None

Answer Key - Topic 9: Puzzles

1: (B) | 2: (A) | 3: (B) | 4: (B) | 5: (B) | 6: (B) | 7: (A) | 8: (B) | 9:
(A) | 10: (B)
11: (C) | 12: (D) | 13: (C) | 14: (C) | 15: (B) | 16: (A) | 17: (C) | 18: (B)
| 19: (A) | 20: (A)
21: (B) | 22: (A) | 23: (A) | 24: (B) | 25: (A)

Topic 10: Clocks & Calendars

1. Theory & Comprehensive Concepts

Clock and Calendar problems test mathematical reasoning involving time cycles, angle calculations between clock hands, and day-of-the-week deductions using odd days.

Core Formulas:

- **Clock Angle:** $\theta = |30H - 5.5M|$ where H is hours and M is minutes.
- **Calendar Odd Days:** Ordinary year has 1 odd day; Leap year has 2 odd days. Century years divisible by 400 have 0 odd days.

Solved Example: What is the angle between the hour hand and minute hand at 3:30?

Solution: $\theta = |30(3) - 5.5(30)| = |90 - 165| = 75^\circ$.

Answer: 75 degrees

Practice Questions (25 Questions)

1. What is the angle between clock hands at 3:30?

- | | |
|----------------|----------------|
| (A) 75° | (B) 90° |
| (C) 60° | (D) 45° |

2. How many odd days are there in 100 ordinary years?

- | | |
|-------|-------|
| (A) 5 | (B) 3 |
| (C) 1 | (D) 0 |

3. What is the angle between hands at 4:20?

- | | |
|----------------|----------------|
| (A) 10° | (B) 20° |
| (C) 0° | (D) 15° |

4. If Jan 1, 2004 was Thursday, what day was Jan 1, 2005?

- | | |
|------------|--------------|
| (A) Friday | (B) Saturday |
| (C) Sunday | (D) Monday |

5. How many times do clock hands coincide in 12 hours?

- | | |
|--------|--------|
| (A) 11 | (B) 12 |
| (C) 22 | (D) 24 |

6. What is the angle between hands at 9:00?

- | | |
|-----------------|-----------------|
| (A) 90° | (B) 180° |
| (C) 270° | (D) 0° |

7. Which of the following is a leap year?

- | | |
|----------|----------|
| (A) 1900 | (B) 2000 |
| (C) 2100 | (D) 2200 |

8. Angle between hands at 6:00?

- | | |
|-----------------|-----------------|
| (A) 0° | (B) 90° |
| (C) 180° | (D) 360° |

9. If today is Monday, what day will it be after 61 days?

- | | |
|--------------|---------------|
| (A) Saturday | (B) Sunday |
| (C) Tuesday | (D) Wednesday |

10. How many times are clock hands at right angles in 24 hours?

- | | |
|--------|--------|
| (A) 22 | (B) 24 |
| (C) 44 | (D) 48 |

11. What day of the week was Jan 1, 2006 if Jan 1, 2005 was Saturday?

- | | |
|-------------|--------------|
| (A) Sunday | (B) Monday |
| (C) Tuesday | (D) Saturday |

12. Angle between hands at 8:20?

- | | |
|-----------------|-----------------|
| (A) 130° | (B) 120° |
| (C) 140° | (D) 150° |

13. How many leap days occur in 400 years?

- | | |
|---------|--------|
| (A) 97 | (B) 96 |
| (C) 100 | (D) 95 |

14. Angle between hands at 2:30?
- (A) 105° (B) 95°
(C) 115° (D) 125°
15. If 15th August is Friday, what day is 31st August same year?
- (A) Saturday (B) Sunday
(C) Monday (D) Friday
16. How many times do clock hands point in opposite directions in 12 hours?
- (A) 11 (B) 12
(C) 22 (D) 24
17. What is the angle at 10:10?
- (A) 115° (B) 120°
(C) 125° (D) 130°
18. If Republic Day 2010 was Tuesday, what day was Independence Day 2010?
- (A) Sunday (B) Monday
(C) Tuesday (D) Wednesday
19. Angle between hands at 5:15?
- (A) 67.5° (B) 75°
(C) 82.5° (D) 90°
20. How many odd days in 400 years?
- (A) 0 (B) 1
(C) 3 (D) 5
21. What is the angle at 7:36?
- (A) 18° (B) 24°
(C) 30° (D) 36°
22. If March 5, 2011 was Saturday, what was March 5, 2012?
- (A) Saturday (B) Sunday
(C) Monday (D) Tuesday

23. Angle between hands at 1:10?

(A) 25°

(B) 30°

(C) 35°

(D) 40°

24. Calendar for which year will be identical to 2005?

(A) 2010

(B) 2011

(C) 2012

(D) 2016

25. How many hours in a week?

(A) 168

(B) 144

(C) 192

(D) 150

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